

VAISHNAVI R

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Summary

Mathematics and Physics student pursuing Data Science, with research experience developing machine learning and data-driven models for complex real-world problems. Combines strong foundations in mathematical modeling, scientific computing, machine learning, and rigorous model evaluation with hands-on experience taking research ideas from problem formulation to implementation and experimentation.

Experience

ML Research Intern

May 2026 – July 2026

Indian Institute of Technology Bombay

- Developed a hybrid physics-ML PMSM Digital Twin combining dq-axis physics, RK4 simulation, PSO calibration, and neural residual correction, reducing held-out current-prediction RMSE by 86.7%, from 11.61 A to 1.55 A
- Built anomaly-detection, severity-estimation, and temperature-compensated degradation-monitoring pipelines, achieving 95.1% ITSC detection on measured fault batches and 93.2% detection of permanent-magnet flux-loss cases
- First author, *A Physics-Informed Hybrid Digital Twin for Scalable Health Management of PMSMs* - manuscript submitted to *IEEE Transactions on Industrial Informatics*; co-authored with Dr. Vibhor Pandhare, Assistant Professor, IIT Bombay

Research Intern – Computational Mathematics Project

Aug 2025 – Dec 2025

Mentored by PhD Scholar, IIT Madras

- Studied the mathematical foundations of the Black-Box Fast Multipole Method for accelerating large-scale kernel computations
- Implemented a one-dimensional BBFMM solver in Python and benchmarked its scaling against direct dense kernel evaluation

Education

B.S. in Data Science and Applications

Expected 2027

Indian Institute of Technology Madras (Online)

B.Sc. (Hons.) in Mathematics and Physics

Expected 2027

Christ University, Bangalore

Projects

Physics-Informed Weak-SINDy Thermal Digital Twin for PMSM

GitHub

Python, NumPy, SciPy, PyTorch, SINDy, RK4

- Developed an energy-conserving weak-SINDy framework to discover missing thermal dynamics in a physics-based PMSM model using 1.33M measurements across 184.8 operating hours, avoiding direct differentiation of noisy temperature data
- Improved unseen-profile temperature RMSE by 53.7%, from 7.61°C to 3.52°C, and permanent-magnet temperature RMSE by 69.6%, from 13.75°C to 4.18°C across 14 untouched test profiles
- Designed a leakage-free profile-level evaluation and sparse model-selection pipeline using 189,315 weak-form equations, identifying interpretable speed-, load-, and loss-dependent mechanisms while enforcing energy conservation

Adaptive Solver Selection Platform for Dynamical Systems

GitHub | Project Link

Python, SciPy, PyTorch, scikit-learn, Streamlit, Scientific ML

- Benchmarked 10+ classical, hybrid ML, physics-informed, and data-driven methods using held-out error, runtime, and residual diagnostics
- Built automated model recommendations for chaotic, stiff, and model-mismatch dynamics using known equations or uploaded time-series data
- Validated selected methods across benchmark systems, including a hybrid RK4 and residual neural network for misspecified dynamics

Technical Skills

Languages: Python, SQL, Java

Machine Learning: PyTorch, scikit-learn, Deep Learning, Residual Learning, Anomaly Detection

Scientific Computing: NumPy, SciPy, Pandas, Numerical ODE Solvers, Mathematical Modeling

Methods: SINDy, System Identification, Physics-Informed ML, Scientific ML, Time-Series Modeling, Model Evaluation

Tools: Git, GitHub, Streamlit